(a)

Problem Settings View

PINNStudio Examples: Forward | Helmholtz 2D – sound-hard scatteri | Default — Open | Save | Export script | Solve | Stop

MODEL
- Problem
- Equations
- Geometry
- Collocation points
- Boundary conditions
  - BC 1 · Neumann, u...
  - BC 2 · Neumann, u...
  - BC 3 · Robin, u_re
  - BC 4 · Robin, u_im
- Neural network
- Training
  - Loss weights
  - Adaptive training
- Parameter sweep
- Results
  - Restore & visualize

SETTINGS

BC 3 · Robin, u_re

One boundary condition: its type, the output it applies to, where on the boundary (Where:) and its value.

Condition

Type: Robin BC — Output: 1

Where: Custom expression... — left or right or bottom or top

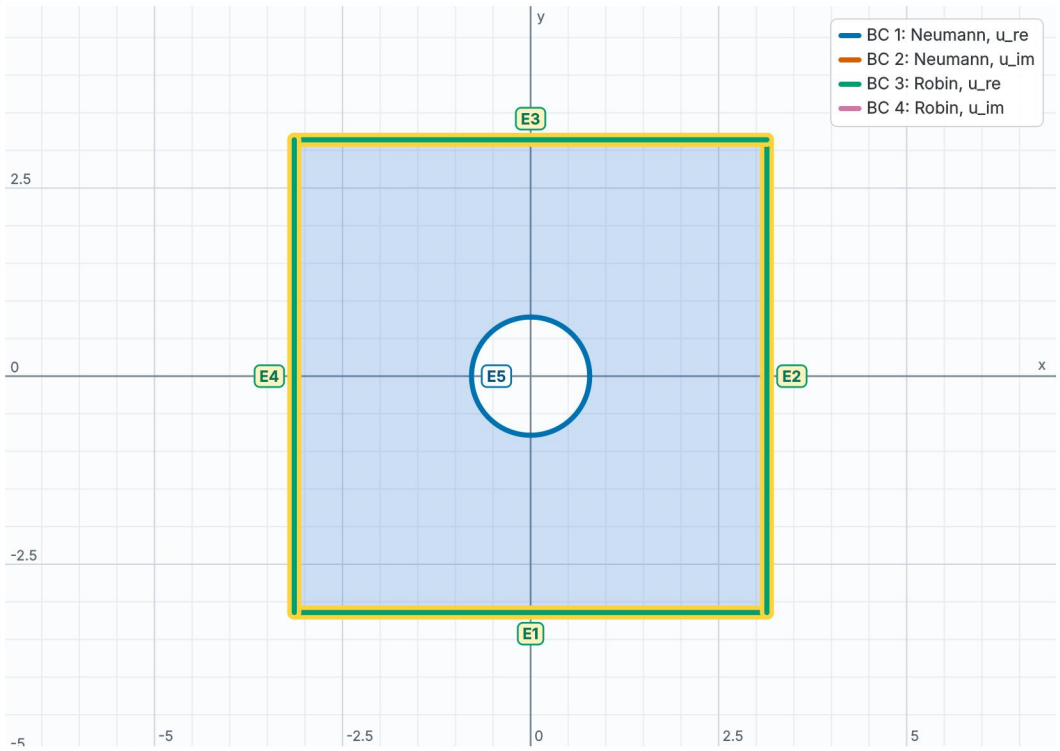
Value: -k0*u_im

On the domain

Picks E1, E2, E3, E4. Click edges in the graphics to add them to this BC or take them out.

Start | Results | Geometry

Fit view



Editing BC 3: click an edge to add it to this BC, or click one of its edges to take it out.

Training Log | Script

Training log will appear here...

Ready — float32 · Save to: ~/PINNStudio_Results

(b)

Problem Settings View

PINNStudio Examples: Forward | Poisson 3D – cube with spherical ci | Default — Open | Save | Export script | Solve | Stop

MODEL
- Problem
- Equations
- Geometry
- Collocation points
- Boundary conditions
  - BC 1 · Dirichlet, u
  - BC 2 · Dirichlet, u
- Neural network
- Training
  - Loss weights
  - Adaptive training
- Parameter sweep
- Results
  - Restore & visualize

SETTINGS

BC 2 · Dirichlet, u

One boundary condition: its type, the output it applies to, where on the boundary (Where:) and its value.

Condition

Type: Dirichlet BC — Output: 1

Where: Custom expression... — np.isclose(x**2 + y**2 + (z - 1)*

Value: sin(pi*x)*sin(pi*y)*sin(pi*z)

On the domain

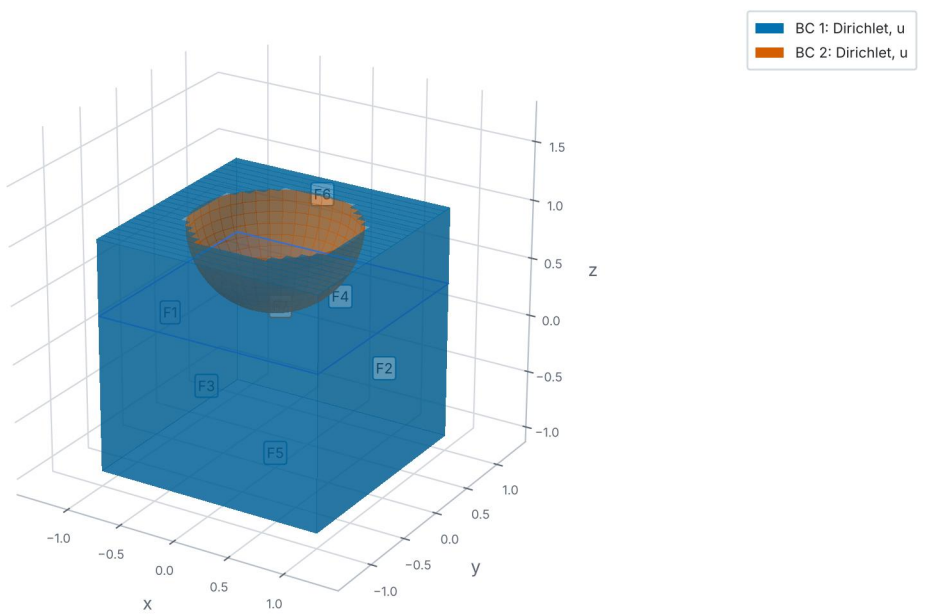
Picks F7. Click faces in the graphics to add them to this BC or take them out.

Start | Results | Geometry

Fit view

View: 3D — Slice at z = 0.350

Iso | Top | Front | Side | Show points



Drag to rotate, scroll to zoom. Each face takes the colour of its boundary condition; the translucent plane is the slice the flat views show. Draw shapes and pick faces in a flat view, or pick faces in the list below.

Training Log | Script

Training log will appear here...

Ready — float32 · Save to: ~/PINNStudio_Results